



Exhaustive matching of 3D/2D coronary artery structure based on imperfect segmentations

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Abstract

Purpose The 3D/2D coronary artery registration technique has been developed for the guidance of the percutaneous coronary intervention. It introduces the absent 3D structural information by fusing the pre-operative computed tomography angiography (CTA) volume with the intra-operative X-ray coronary angiography (XCA) image. To conduct the registration, an accurate matching of the coronary artery structures extracted from the two imaging modalities is an essential step.

Methods In this study, we propose an exhaustive matching algorithm to solve this problem. First, by recognizing the fake bifurcations in the XCA image caused by projection and concatenating the fractured centerline fragments, the original XCA topological structure is restored. Then, the vessel segments in the two imaging modalities are removed orderly, which generates all the potential structures to simulate the imperfect segmentation results. Finally, the CTA and XCA structures are compared pairwise, and the matching result is obtained by searching for the structure pair with the minimum similarity score.

Results The experiments were conducted based on a clinical dataset collected from 46 patients and comprising of 240 CTA/XCA data pairs. And the results show that the proposed method is very effective, which achieves an accuracy of 0.960 for recognizing the fake bifurcations in the XCA image and an accuracy of 0.896 for matching the CTA/XCA vascular structures.

Conclusion The proposed exhaustive structure matching algorithm is simple and straightforward without any impractical assumption or time-consuming computations. With this method, the influence of the imperfect segmentations is eliminated and the accurate matching could be achieved efficiently. This lays a good foundation for the subsequent 3D/2D coronary artery registration task.

Keywords 3D/2D coronary artery · CTA/XCA structure matching · Exhaustive matching algorithm · Imperfect segmentations

Introduction

Currently in clinical practice, the percutaneous coronary intervention [1] is routinely performed for the treatment of coronary artery disease [2] under the guidance of X-ray coronary angiography (XCA) image due to its advantages

in real-time and multi-view observation. However, the projective nature of XCA image causes the loss of spatial information and further creates ambiguities for the interpretation of the original 3D coronary artery structures, thus hampering the intervention. This issue could be addressed by conducting the 3D/2D coronary artery registration [3–7], which fuses the vascular structure extracted from the pre-operative 3D computed tomography angiography (CTA) volume with that extracted from the intra-operative 2D XCA image to achieve better guidance. The state-of-the-art coronary artery registration algorithms are commonly feature-based, which apply the centerlines as the registration target since they preserve the essential topological information of the vascular structures.

The coronary artery is a treelike structure consisting of several vessel segments. Therefore, an accurate matching of each vessel segment across the two imaging modalities is

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the prerequisite to conduct the registration. However, there are some problems which make this task challenging. First, the extracted coronary arteries are from different physical dimensions and there are significant shape differences existing in each vessel segment caused by the heartbeat and the respiration. Second, the structure overlapping in XCA image generates fake bifurcations, which makes it difficult to obtain the real coronary artery topological structure and further misleads the matching. Last but not least, the extracted vascular structures that are theoretically identical in the two imaging modalities might actually be different, since the performances of the segmentation algorithms [8–11] are rarely perfect. Specifically, the thin vessel segments might be missing.

Despite these difficulties, some literature has focused on this matching task and made certain achievements [12–15]. However, they still have limitations. First, the computations in the existing works are usually complex, which is time-consuming. Second, the real bifurcations and the fake bifurcations are equally treated in XCA image, which further increases the structure complexity and the computation time. Third, the existing works are based on the assumption that each vessel segment in the CTA volume should have a corresponding vessel segment in the XCA image. Considering the uncertainties of the segmentation results, this assumption is actually not reasonable.

To overcome these limitations, in this study we propose a simple and fast algorithm, which does not require complex computations, for the 3D/2D coronary artery structure matching task. First, the fake bifurcations in the XCA image are recognized and the fractured centerline fragments are concatenated. This restores the original vascular structure and will avoid the unnecessary computations caused by the fake bifurcations in the subsequent matching step. Then, considering that the extracted vascular structures which are theoretically identical in the two imaging modalities are usually different due to the limited segmentation performance, the vessel segments in the two imaging modalities are removed orderly in an exhaustive way to simulate all the possible imperfect segmentation results. The generated CTA and XCA structures will be compared one by one to select the data pair with identical topological structure as candidates, and the final matching result is obtained by searching for the candidate pair with the minimum similarity score. Note that the vessel segments of the two imaging modalities are treated equally in the matching process without any assumption.

Related works

The existing works usually regards the vascular structure matching task as a graph matching problem, where the bifur-

cations and the ending points of the vessel segments are regarded as nodes. The CTA centerline structure and the XCA centerline structure are regarded as a reference model and a redundant graph model, respectively. The target is to find a subgraph from the redundant graph. Benseghir et al. [16] extended the iterative closet point algorithm to the iterative closet curve algorithm by using the vessel segment as the pairing element instead of point and measured the similarity of two centerline curves with Fréchet distance. On this basis, the authors [12] further imposed consistency to the bifurcation points to ensure a one-to-one matching of the segment curves. And the problem of finding the optimal tree pairing is broken into subproblems of finding an optimal pairing for each child subtree, which is solved by the recursive algorithm. Liu et al. [14] improved this matching strategy from one-to-one node to many-to-many nodes under the constraints of tree topology. Different from [12], the fake bifurcations have been eliminated in advance to make the XCA vessel tree a directed acyclic graph. Two adjacent CTA nodes will be matched with two XCA nodes and combined with multiple nodes on the path between them. And all the potential matching results are considered. Zhu et al. [15, 17] adopted a coarse-to-fine matching strategy. In the coarse stage, the nodes including the fake bifurcations are matched with a redundant graph matching method. The node affinity matrix and edge affinity matrix are generated to form the score function, which is solved as a quadratic assignment problem. In the fine stage, the vessel points are matched by assembling each edge curve matching result. Different from the literatures above, Chen et al. [13] proposed a novel method based on circuit simulation, where the circuit conversion model is applied to simulate the vascular graph. The nodes were matched by analyzing the corresponding voltage response.

Although these methods have made certain achievements, they still have limitations. First, all the works are based on the potential assumption that each vessel segment in the CTA volume should have a matching element in the XCA image. However, due to the complexities of the CTA volume and the XCA image, the current segmentation algorithms still cannot guarantee a perfect extraction of all the vessel segments, which makes the vascular structures of the two modalities not identical. Therefore, this assumption is impractical. Second, some of the works [12, 15–17] do not involve the recognition of the fake bifurcations, which are equally considered as nodes in the vessel graph as the real bifurcations. This makes the graph more complicated and increased the difficulty of the matching task. Third, solving the matching problem usually involves complex computations, for example, the recursive algorithm [12] and the quadratic assignment problem [15, 17]. These computations are time-consuming and not suitable for real-time tasks.

Contributions

The contributions of this study are two folds. First, we propose a method for the 2D topological structure extraction in the XCA image. With the designed rules, the centerline fragments generated by the fake bifurcations are recognized and the centerlines are restored. Second, we proposed a method for the structure matching across the 3D CTA and 2D XCA imaging modalities. With the exhaustive algorithm which generates all the possible structures, the influence of the imperfect segmentation is eliminated and accurate matching is achieved.

Materials and methods

Data description

The raw clinical data were collected from 46 patients at the Department of Cardiothoracic Surgery, Peking Union Medical College Hospital of China. For each patient, a pre-operative CTA volume and about 5 or 6 intra-operative XCA sequences based on different projection parameters were collected. The CTA volumes were acquired on the Siemens SOMATOM Definition Flash CT platform. The number of slices ranges from 210 to 380 and the resolution of each slice is 512×512 . The voxel size is $0.35 \times 0.35 \times 0.75$ (unit: mm). The XCA sequences were acquired on Philips INTEGRIS Allura Flat Detector. The frame rate is 15 fps. The number of frames ranges from 40 to 75 and the resolution of each frame is 512×512 . The pixel size is 0.37×0.37 (unit: mm). For each XCA sequence, a frame full of contrast agent is selected by the cardiologist to form a data pair with the corresponding CTA volume. There are totally 240 data pairs, where 156 are left coronary arteries (LCA) and 84 are right coronary arteries (RCA). And there are totally 2165 CTA/XCA vessel segment pairs in the whole dataset and 249 fake bifurcation fragments in all the XCA images.

Methods

The whole method consists of two main steps as shown in Fig. 1. In the pre-processing step, the coronary artery structures are first extracted from the two imaging modalities based on the methods [18, 19] we previously developed. Then, the centerlines are extracted with the morphological operations. The CTA centerlines go through a coordinate transformation and are projected to the XCA image plane based on the parameters in the DICOM head file [20]. And a translation is conducted to shorten the distance between the projected CTA centerlines and the XCA centerlines: first, the root points of the vascular trees are recognized by selecting

the bifurcations with the maximum radius. Then, the original 3D CTA centerlines are moved to align the roots of the two modalities. After this pre-alignment, the structure matching is conducted, which is the core of this study. First, the fake bifurcations generated by the overlapping of the vascular structures in the XCA image will be recognized and the damaged centerline structures will be restored. Then, the vessel segments in the CTA and XCA imaging modalities will be removed orderly to simulate the imperfect vessel segmentation results. All the possible centerline structures will be generated exhaustively and matched to find out the best matching result. The following sections will describe them in detail.

Fake bifurcations recognition and centerline restoration

Due to the projective imaging nature, the tubular coronary artery structures overlap in the XCA image and generate small overlapping areas called the fake bifurcations as shown in Fig. 2. When extracting centerlines, these fake bifurcations also produce small pieces of centerlines, which divide the complete centerline of a vessel segment into several fragments. This greatly damages the original topological information that is essential for the coronary artery structure matching task. In this section, we propose a method, which consists of two steps, to solve this problem.

In the first step, the raw centerline fragments that are continuous pixels in the binary image are extracted and transformed to arrays recording the coordinates of each pixel in the image space. A centerline fragment is defined as $C_f = [p_0, p_1, \dots, p_n] \in R^{2 \times n}$, where $p_n = [x_n, y_n]^T$ (x_n and y_n denote the line and column numbers, respectively). And considering that the fake bifurcations are generated by the intersections of two vessel segments, the related centerline fragments should have four neighboring fragments and relatively short length (smaller than 20 pixels according to our experience) as shown in Fig. 2. Based on these two criteria, the fake bifurcation fragments are selected from the extracted fragments.

In the second step, the original centerline structure is restored by eliminating the fake bifurcation fragments, which are connected with the upstream fragments and corresponding downstream fragments. Starting from the root point of the coronary artery which is recognized by searching for the position with the maximum diameter, we traverse the centerline fragments pixel by pixel in the direction the vessel extends to simulate the direction of the blood flow as shown in Fig. 3. Since the coronary artery is a tree structure with several pathways growing from the root, this process is conducted from different directions simultaneously. And according to different blood flow situations, we summarize three types of fake bifurcation fragments with respective processing methods. In the first situation (shown in Fig. 3a), the blood flow in

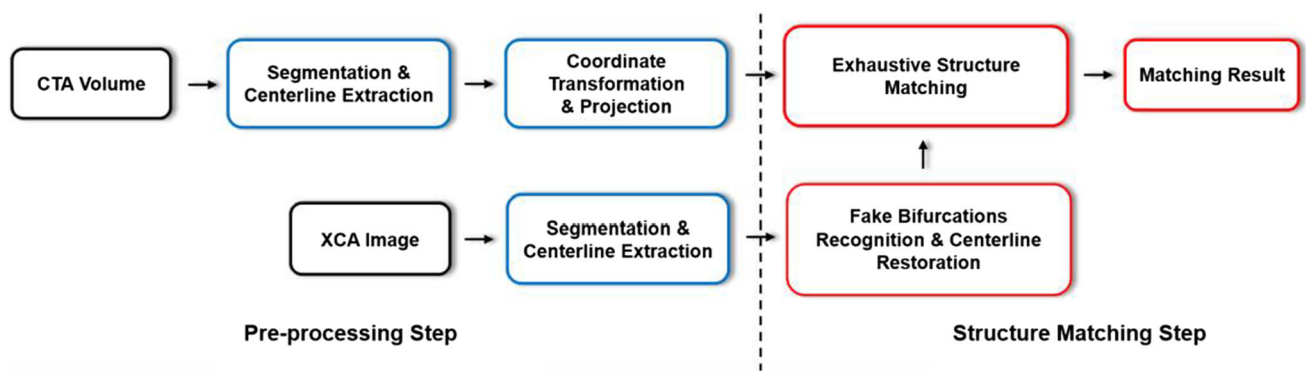


Fig. 1 The workflow of the proposed method

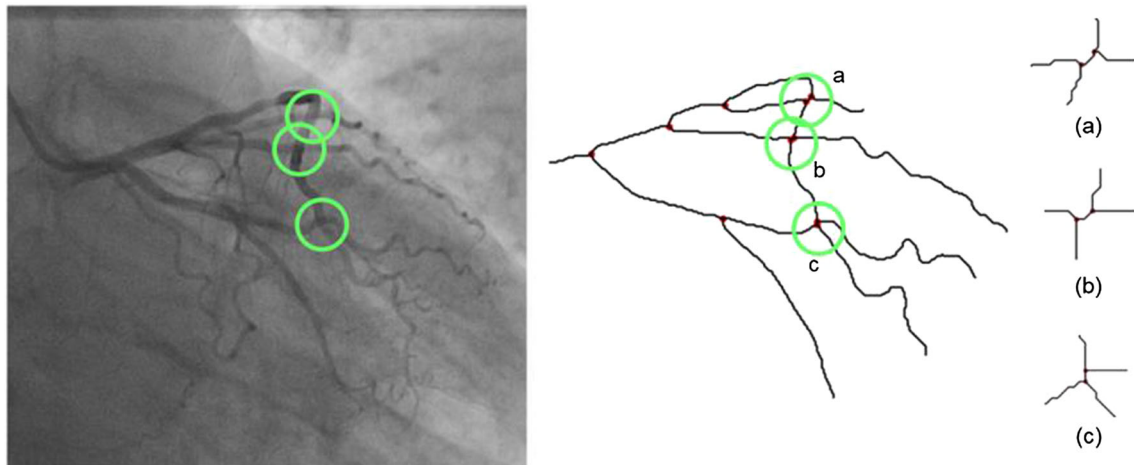
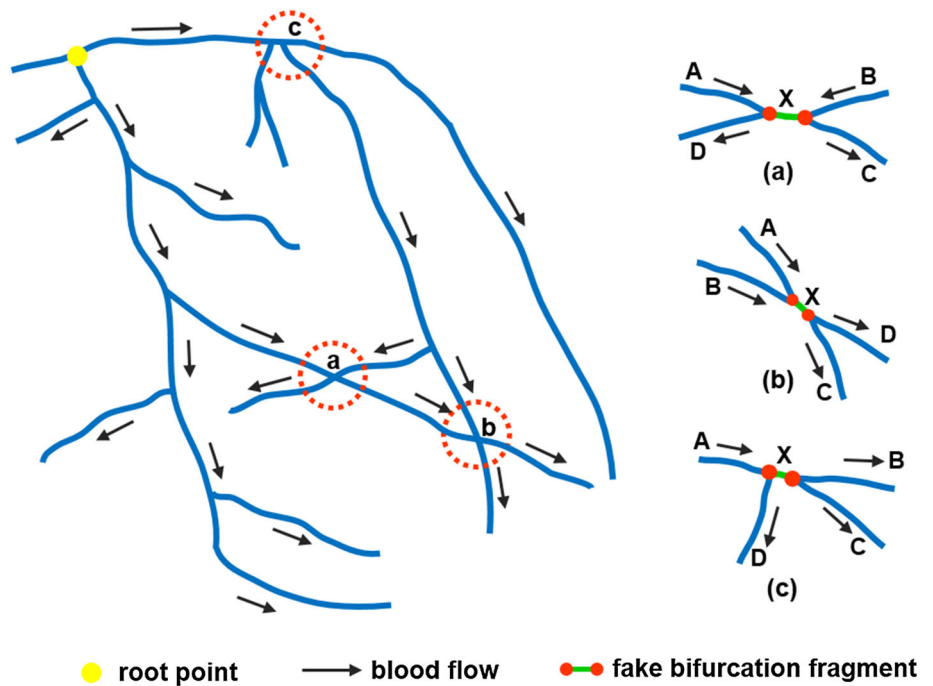


Fig. 2 The raw XCA image (left) and the extracted centerline structure (right). The fake bifurcations and corresponding centerline fragments are highlighted in green circle

Fig. 3 The coronary artery blood flow pathway (left) and three types of fake bifurcation fragments (right): **a** the blood flow comes from two directions; **b** the blood flow comes from the same direction; **c** the situation that requires further discrimination



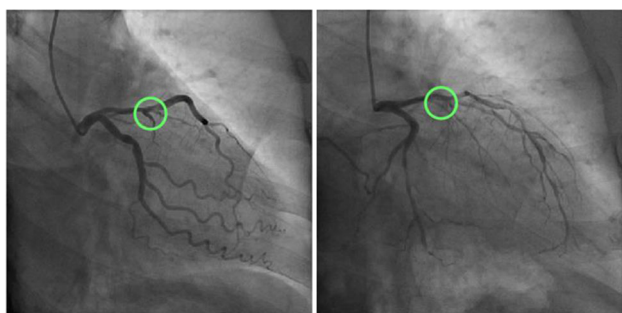


Fig. 4 Two fragment samples of the third type (highlighted with green circles). Left: **a** fake bifurcation fragment. Right: **b** real short vessel segment

the fake bifurcation fragment that comes from two upstream fragments has two different directions. Therefore, to maintain the consistency of the blood flow directions, it's obvious that the restored centerline segments should be (A, X, C) and (B, X, D) . In the second situation (shown in Fig. 3b), the blood flows of the two upstream fragments have identical directions and further analysis is required. Considering that the vessel segments are smooth and their extending directions change slightly in the local area, the fragments could be matched according to the differences between the slopes of the two upstream fragments and two downstream fragments. Since $|k_A - k_C| + |k_B - k_D| < |k_A - k_D| + |k_B - k_C|$ (where k_* is the slope of each fragment), the restored centerline segments should be (A, X, C) and (B, X, D) . The third situation might be the most complex (shown in Fig. 3c). The fragment X only has one upstream fragment. It might be a real short vessel segment or a fake bifurcation fragment generated by the overlapping of vessel segment B and vessel segment C (as shown in Fig. 4). The processing of this type of fragment would be introduced in the next section.

Exhaustive structure matching

After the fake bifurcation fragments are recognized and the related centerlines are restored, the coronary artery structures in the XCA modality will be matched with those in the CTA modality. Theoretically, the coronary artery structures in the two modalities should be identical since they are collected from the same patient. However, since the performances of the applied vessel segmentation algorithms are limited by lots of factors (e.g., the background noise, the concentration of contrast agent and the blood flow), the extracted vascular structures are rarely complete (some vessel segments are missing) and might be locally different between the two imaging modalities. For example, a vessel segment might be successfully extracted from the CTA volume and fails to be extracted from the XCA image, and vice versa. Therefore, the vascular structures of the two imaging modalities are actually

inconsistent and the matching of the missing vessel segments is theoretically impossible. The best solution for this problem might be developing segmentation algorithms with better performance and robustness. However, this is not the focus of this study. Considering that the imperfect segmentation result is common and hard to avoid, in this study we focus on establishing correspondence for the obtained vascular structures regardless of the missing parts. If a certain vessel segment is successfully extracted only in one imaging modality without corresponding result in the other imaging modality, this vessel segment will be removed from the matching task and the original topological structure of the coronary artery will be reorganized.

To simulate the imperfect segmentation results, the proposed method will generate all the potential structures exhaustively (as shown in Fig. 5). Note that in a coronary artery tree, the upstream vessel segments that are close to the root point usually have relatively big diameters and could be easily extracted. And on the contrary, the ending vessel segments are usually long and thin, which have greater possibilities of being ignored by the applied segmentation algorithms. Therefore, the distal thin vessel segments will be considered to have the possibilities of absence. And some rules are designed for the construction of the new topological structure (as shown in Fig. 6). First, if a vessel segment is removed, its parent segment and sibling segment will form a new vessel segment. Second, if a vessel segment is removed, all the child segments will be removed simultaneously. Finally, the third type of fake bifurcation fragments (as introduced in “[Fake bifurcations recognition and centerline restoration](#)” section) are also handled in this section since they also have the chance of absence. In this situation, the parent segment will form two new segments with the child segments, respectively.

Based on the idea of exhaustion and the designed rules, all the possible CTA and XCA vascular structures are generated, which are abstracted into topological structures (as shown in Fig. 5). And all the CTA topological structures and all the XCA topological structures will be compared one by one to select the data pair with identical structures. Similarity measurement is then conducted on these candidate CTA/XCA pairs to find the best matching pair, which works as follows: first, the CTA centerlines are projected to the XCA image plane according to the projection method [20] and the angle parameters recorded in the DICOM header file of the corresponding XCA image. Then, the dynamic time warping (DTW) distance [21] between each CTA projected vessel segment and its matched XCA vessel segment is calculated. DTW algorithm could establish correspondence for the points of two sequences with unequal length and calculate the accumulated distance. And the distance between a matched point pair is measured with Euclidean distance. Supposing

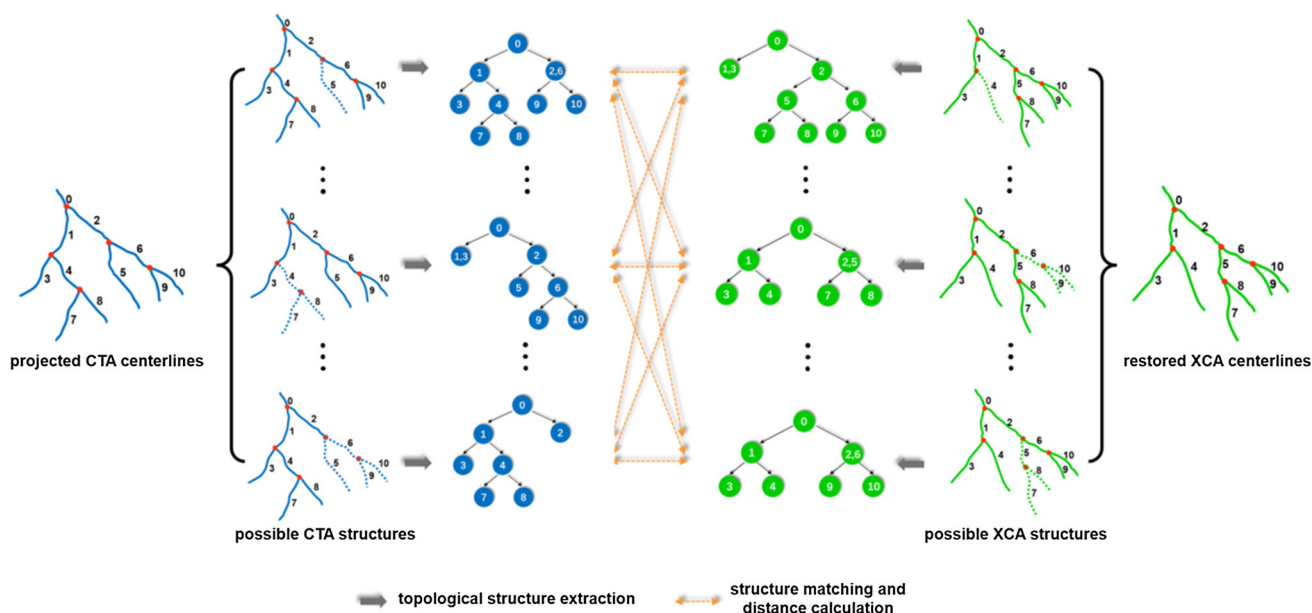


Fig. 5 The process of the proposed exhaustive matching algorithm. The possible CTA and XCA vascular structures are generated and abstracted into topological structures. The topological structures of CTA and XCA

are compared one by one to select the identical pairs. Similarity measurement is then conducted on these candidate CTA/XCA pairs to find the best matching result

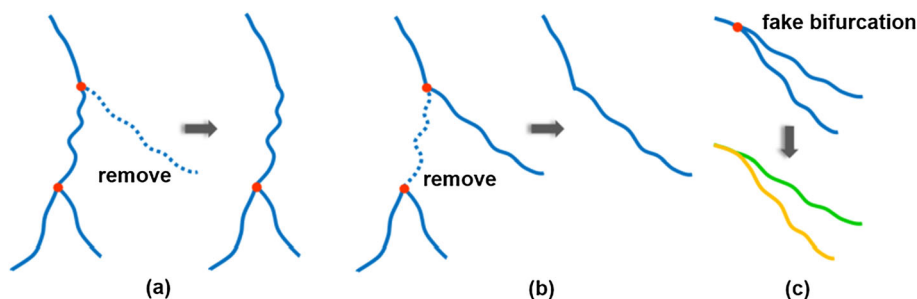


Fig. 6 The rules for reorganizing the new vascular structure. **a** The parent segment and sibling segment of the removed vessel segment forms a new vessel segment; **b** the child segments of the removed vessel segment are removed; **c** the parent segment forms two new segments with

the child segments when dealing with the potential fake bifurcation fragment

there are n vessel segment pairs in a matched CTA/XCA structure, the overall similarity score s will be computed as:

$$s = \sum_{i=1}^n \text{DTW}(c_i, x_i) / \left(\sum_{i=1}^n l_i * n \right) \quad (1)$$

where $\text{DTW}(c_i, x_i)$ denotes the DTW distance between a projected CTA vessel segment centerline c_i and its matched XCA vessel segment centerline x_i . And l_i denotes the length (the number of the voxels) of a CTA vessel segment. Note that the n in the denominator serves as a penalty term. By dividing n in the distance metric, the CTA/XCA structure pair with relatively more vessel segments will have relatively smaller distance score, which leads the matching algorithm to choose the matching result that more completely preserves

the original coronary artery structures. The final matching result is determined by finding the CTA/XCA structure pair with the minimum similarity score.

Experiments and results

Implementation details

The proposed method is conducted based on a 4-core 2.4 GHz Intel Xeon E5-2630 v3 processor and operating system Ubuntu 20.04 LTS. The code is implemented with Python 3.6.7. Based on these experimental settings, the average computation time is 0.879 s for one coronary artery sample, where the fake bifurcation recognition and centerline restoration in

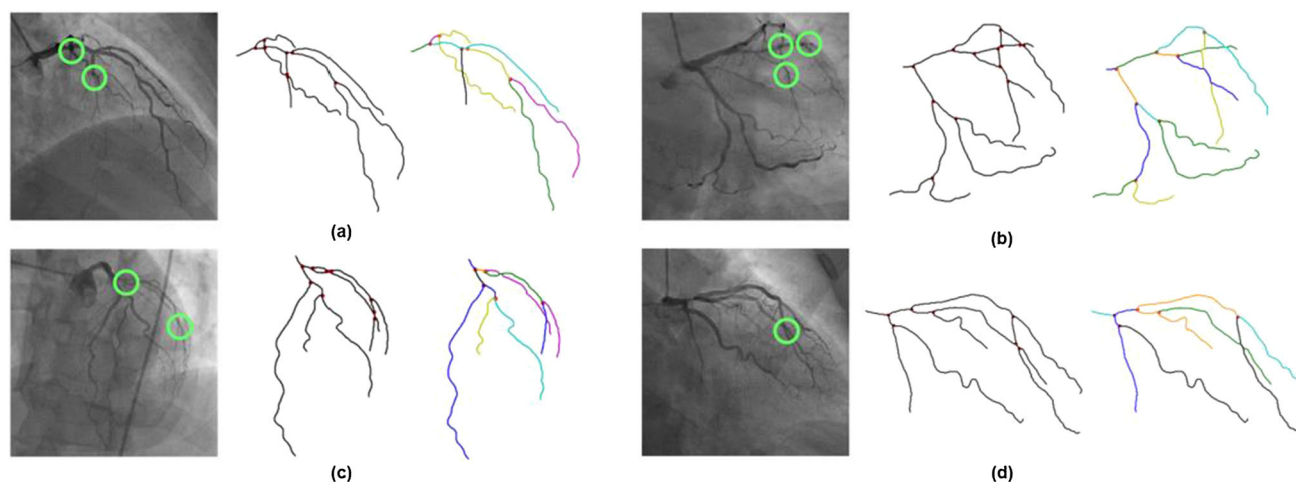


Fig. 7 The results of four XCA image samples before and after the centerline structure restoration. For each sample, the raw image (fake bifurcations are highlighted in green circle), the raw centerline structure, and the restored centerline structure are shown, respectively. And

the vessel segments of the restored centerline structures are shown in different colors

Table 1 The vessel segment matching accuracies of LCA and RCA based on two criteria: sample-level: the accuracy is measured based on all the CTA/XCA data pairs; segment-level: the accuracy is measured based on all the vessel segment pairs

Structure	Accuracy	
	Sample-level	Segment-level
LCA	0.872	0.954
RCA	0.940	0.974
All	0.896	0.960

the XCA image takes up 0.794 s and the exhaustive matching step takes up 0.085 s.

The performance of the proposed algorithm is evaluated with the metric accuracy (ACC), which is computed as follows: $ACC = a/b$. When conducting the centerline restoration experiment, a denotes the number of successful restorations (a fake bifurcation fragment is connected with the correct upstream and downstream fragments) and b denotes the number of all the fake bifurcation fragments. When conducting the structure matching experiment, ACC is measured from both sample-level and segment-level. In sample-level, a denotes the number of the CTA/XCA data pairs whose vessel segments are all correctly matched and b denotes the number of all the CTA/XCA data pairs. In segment level, a denotes the number of the correctly matched CTA/XCA vessel segment pairs and b denotes the number of all the CTA/XCA vessel segment pairs.

The centerline restoration experiment

In this section, experiments are conducted to recognize the fake bifurcation fragments in the XCA centerline image and restore the original topological structures. Figure 7 shows some samples after the centerline restoration. It can be observed that the fake bifurcations fragments have been concatenated with the correct upstream fragments and downstream fragments. Based on the 249 fake bifurcation fragments in the dataset, the restoration accuracy is 0.960.

The exhaustive structure matching experiment

In this section, the exhaustive vascular structure matching experiments are conducted. In a vascular tree, the middle vessel segments that are relatively thick with more contrast agent are usually extracted successfully. And the terminal vessel segments that are thin and far from the inlet of the coronary artery have greater possibilities to be ignored by the segmentation algorithms. By observing the actual performances of the segmentation algorithms on the applied clinical dataset, we notice that in the worst situation there are four vessel segments that fail to be extracted. Therefore, when considering the number of possible CTA and XCA coronary artery structures, the maximum number of the missing vessel segments is set to 4, which generates 16 (2^4) possible CTA structures and 16 (2^4) possible XCA structures, respectively.

The final matching result is obtained by exhaustive matching between these potential structures. The matching accuracy, which is measured based on two criteria (sample-level and segment-level), is shown in Table 1. And the accuracies

of the left coronary artery (LCA) and right coronary artery (RCA) are listed, respectively. Since the RCA structures are relatively simple with less vessel segments than the LCA structures and the number of fake bifurcations is very small, the proposed method shows better performance for matching the RCA structures.

Discussions and conclusions

In this study, we propose a new method to match the coronary artery structures between the CTA imaging modality and the XCA imaging modality, which will serve as a pre-processing step for the 3D/2D coronary artery registration task. The proposed method is based on the 2D topological structure restoration and the exhaustive structure matching, which is simple and straightforward without any impractical assumption or time-consuming computations as applied in the existing works. And a matching accuracy of 0.896 is achieved on the collected clinical dataset. Despite the good performance, we still believe there are some problems worth discussing.

When conducting the coronary artery matching task in the existing works, the fake bifurcations in the XCA image are not specially treated. The structure matching is conducted as a graph matching task, where the fake bifurcations, real bifurcations and the ending points of vessel segments are equally considered as nodes. However, we do not think this is an appropriate way of treating the fake bifurcations. The existing methods are usually based on complex computations, involving fake bifurcations will increase unnecessary computation and further damage the algorithm efficiency. And as we have introduced in “[Fake bifurcations recognition and centerline restoration](#)” section, the fake bifurcation fragments have some obvious characteristics (small length and four neighboring fragments). Therefore, in this study we choose to recognize these fragments and reduce the complexities for the subsequent matching task.

Another problem worth noticing is the influence of the segmentation algorithms that extract the vascular structures from the raw image. In this study, the applied segmentation algorithms [18, 19] are deep learning-based and developed by our lab and research partners recently with the state-of-the-art performance. However, due to the complexities of the coronary artery structures, the extracted vascular structures are still not perfect. In certain cases, the segmentation algorithm completely ignores the thin vessel segment, which generates an incomplete vascular tree and makes the structures of the two imaging modalities inconsistent. This type of data takes up about 3.0% of the extracted vessel segment pairs. As we have introduced previously, the existing matching algorithms are usually based on the assumption that all the vessel segments in the CTA imaging modality should

have a matching result in the XCA imaging modality. Therefore, when facing a relatively poor XCA segmentation result, sometimes the existing works don't work. Although developing segmentation algorithms with higher performances might be a significant future work to solve this problem, in this study we focus on matching the obtained structures regardless of the missing vessel segments by proposing the exhaustive matching algorithm. All the potential structures will be generated with the designed rules to simulate the imperfect segmentation results. The proposed method greatly reduces the dependence on the pre-processing segmentation algorithm, which will make the whole registration framework more robust. Besides, according to the statistics presented in the existing works [12, 14, 17], it usually takes about ten seconds to complete the matching task for a coronary artery sample. However, based on the regular hardware environment we applied, the proposed algorithm only requires less than one second, which is obviously more efficient.

In the future, we will try to improve this study from two aspects. First, considering the diversity and complexity of the coronary artery structures, we will establish a larger clinical dataset with more variations by collecting data from more hospitals. Second, since the proposed method is basically general and has the potential to be applied to other vascular structures, we will try to extend the proposed method to other organs (e.g., the brain and the liver).

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

Research involving human participants and/or animals This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent This article does not contain patient data.

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